Experimental Analysis of Public Policies

Session 6

Event Studies and Staggered DiD

Etienne Dagorn
University of Lille

Session Objectives

By the end of this session, students should be able to:

- define event time and interpret omitted periods, leads, and lags;
- read an event-study graph as both diagnostic evidence and effect dynamics;
- explain why traditional TWFE event-study coefficients need qualification;
- identify anticipation, dynamic effects, and weak support in long lags;
- explain why staggered adoption complicates TWFE;
- distinguish never-treated, not-yet-treated, and already-treated comparison units;
- define $Y_{it}(g)$ and interpret $ATT(g,t)$;
- describe why modern DiD estimators keep cohort and event time explicit.

Roadmap

From 2x2 DiD To Event Time

Event Time And Traditional TWFE

Staggered Adoption: Who Is Control?

Why Naive TWFE Can Mislead

Group-Time ATT And Aggregation

Practice And Bridge

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From DiD To Dynamics

- Session 5: one treated group, one comparison group, one treatment date.
- Core assumption: the comparison group reveals the treated group's untreated trend.
- New problem: effects may appear slowly, fade, or differ across cohorts.
- New question: who serves as control at each date?

Central question

What changes when treatment timing and treatment effects are heterogeneous?

The New Task Is to Name the Comparison at Each Date

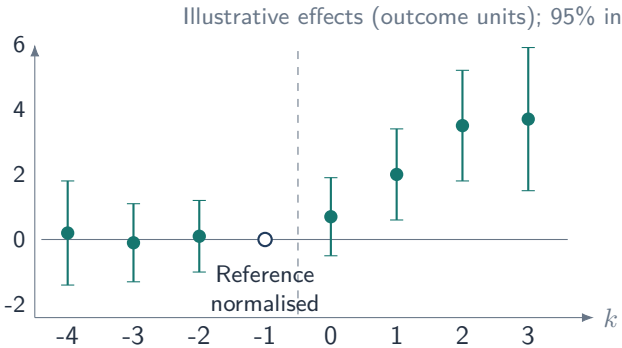
Keep the Session 5 reflex

Which untreated change can stand in for this treated cohort's missing change?

- First align dates and read the graph.
- Then audit control status and contamination.
- Finally define cohort effects and choose an average for the policy question.

Detailed weighting algebra stays in the appendix.

An Event-Study Graph: Describe Before Explaining



1. Find the reference: $k = -1$.
2. Read the pre-policy evidence.
3. Describe when gains appear.
4. Inspect uncertainty and support.

Read the Graph Before Calling It Causal

Use the graph on the preceding slide.

Two minutes in pairs

Name the reference period. Describe one post-policy magnitude and its uncertainty. Can the pre-policy intervals rule out a substantively important violation?

First reading: a common treatment date, an untreated control group, and effects in outcome units. We next examine how staggered dates change that interpretation.

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Event Time

$$k = t - G_i$$

- G_i: first period in which unit i receives treatment.
- $k = 0$: first treated period.
- $k = -1$: period just before treatment.
- $k = 1$: one period after treatment.
- Never-treated units have $G_i = \infty$, so event time is not defined for them.

Event-Time Data Layout

	2022	2023	2024	2025
Early adopter, $G = 2023$	$k = -1$	$k = 0$	$k = 1$	$k = 2$
Late adopter, $G = 2025$	$k = -3$	$k = -2$	$k = -1$	$k = 0$
Never-treated, $G = \infty$	control	control	control	control

Student output

Convert two calendar years into event time and mark which units are untreated in each year.

The Event-Study Equation Behind the Dots

For a common adoption date g, and untreated controls:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \in \mathcal{K},\ k \neq -1} \beta_k \underbrace{1[A_i = 1,\ t - g = k]}_{\text{treated group at event time } k} + u_{it}.$$

- Omit $k = -1$: the period just before adoption is the reference.
- Leads ($k < 0$) describe differential pre-treatment changes.
- Under the design assumptions, lags ($k \geq 0$) describe treated effects over time.

With staggered dates

Replacing g by each unit's G_i is common, but the resulting TWFE dots can mix effects. We will inspect why.

Traditional TWFE Event-Study Regression

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \in \mathcal{K},\, k \neq -1} \beta_k 1[G_i < \infty,\ t - G_i = k] + u_{it}$$

- Unit fixed effects absorb permanent unit differences.
- Time fixed effects absorb common calendar shocks.
- One event-time period is omitted as the reference.
- Never-treated units have all event-time indicators equal to zero.
- Endpoints such as $k \leq -5$ or $k \geq 5$ are often binned.

Traditional TWFE: Immediate Caveat

Do not overread the dots

In staggered designs with heterogeneous effects, traditional TWFE β_k's may not equal clean event-time ATTs.

- Already-treated units can enter comparisons for newly treated units.
- Dynamic effects from early cohorts can contaminate lead or lag coefficients.
- The graph still teaches timing, but it needs a comparison-group audit.

The Omitted Period

- Usually omit $k = -1$, the period just before treatment.
- Each coefficient is interpreted relative to that omitted period.
- Changing the omitted period changes the visual reference point, not the underlying information.

Reading cue

If $k = -1$ is omitted, a coefficient at $k = 2$ is relative to one period before treatment.

Leads And Pre-Trend Diagnostics

- Lead coefficients correspond to periods before treatment.
- In a clean comparison design, leads should be close to zero under no anticipation and parallel pre-trends.
- In traditional staggered TWFE, leads may also reflect contamination from heterogeneous post-treatment effects.
- They are diagnostics, not definitive tests.

Low-power warning

A flat pre-period with wide confidence intervals may simply mean the design cannot detect violations.

Lags And Dynamic Effects

- Lag coefficients correspond to periods after treatment.
- In a clean event-study estimator, lags describe post-treatment dynamics.
- In traditional staggered TWFE, lags need a comparison-group and weighting audit before causal interpretation.
- Long-run lags may be imprecise if few cohorts are observed that long.

> **Policy example**
>
> A training programme may show no effect at $k = 0$, gains at $k = 2$, and fade-out by $k = 5$.

Anticipation

- Units may respond before official treatment starts.
- Firms may invest before a tax change.
- Households may move before a boundary reform.
- Researchers may redefine the treatment date or omit anticipation periods.

Interpretation choice

Sometimes the relevant policy is the announcement plus implementation, not only the legal start date.

How To Read The Graph

1. Locate the omitted period.
2. Identify the estimator and comparison group.
3. Check the pre-treatment coefficients without treating them as proof.
4. Look for a change around treatment onset.
5. Compare magnitudes to policy relevance.
6. Notice confidence intervals and sample support.

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Example: Minimum Wage

- Cities raise minimum wages in different years.
- Employment may adjust slowly.
- Early adopters may differ politically and economically.
- Later adopters can be untreated controls before they adopt, but not after.

Class question

Should a city that raised the minimum wage three years ago serve as a control for a newly treated city?

Staggered Adoption

- Different units adopt the policy in different periods.
- Some units may never adopt.
- Some treated units may act as controls for later-treated units before they adopt.
- Already-treated units may also enter traditional TWFE comparisons.

Why it matters

Control status becomes relative to calendar time and treatment cohort, not a fixed label.

Cohort-By-Calendar Grid

Group	2022	2023	2024	2025
Early adopters	untreated	treated	treated	treated
Late adopters	untreated	untreated	untreated	treated
Never-treated	untreated	untreated	untreated	untreated

Student output

For 2024, mark who is treated, who is not-yet-treated, who is never-treated, and who should not be used as an untreated control.

Staggered Adoption: Debrief Grid

Comparison	Interpretation risk
Treated vs never-treated	clean if never-treated units are valid controls
Treated vs not-yet-treated	useful before later adoption occurs
Treated vs already-treated	risky when effects evolve over time
Early vs late, before late adoption	needs parallel untreated changes and no anticipation

Reflex

In staggered DiD, always ask: who is serving as the control group at each date?

Comparison Groups

- Never-treated units are often the cleanest comparison group if they are credible.
- Not-yet-treated units can be useful before they adopt.
- Already-treated units are problematic if treatment effects evolve over time.
- For a fixed ATT(g,t), choosing controls changes the identifying assumption; limited control support can restrict which targets are identified.

Diagnostic prompt

For each event time, ask whether the comparison units are never-treated, not-yet-treated, or already exposed.

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TWFE Weights: Intuition

- TWFE combines many 2x2 comparisons.
- In staggered designs, some comparisons may be undesirable.
- If effects differ by cohort or time since treatment, the weighted average can be hard to interpret.
- The issue is not fixed effects themselves; it is the comparison structure.

Plain words

A single TWFE coefficient can mix early-vs-late, late-vs-early, and treated-vs-never comparisons.

Why an Already-Treated Control Can Subtract a Real Effect

Schematic example: no untreated trend; treatment effects grow with exposure.

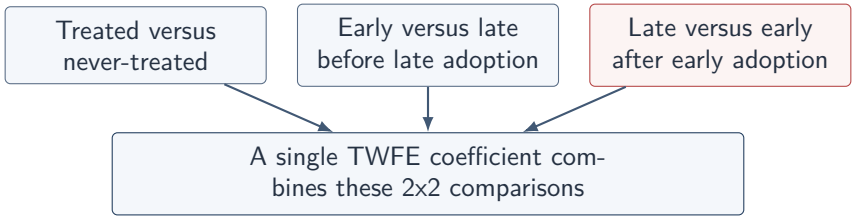
Outcome	Before late adoption	After late adoption	Change
Late adopters	0 (untreated)	2 (newly treated)	$+2$
Early adopters	2 (already treated)	6 (longer exposure)	$+4$
Never-treated	0	0	0

$$DiD_{\text{late vs early}} = 2 - 4 = -2, \qquad DiD_{\text{late vs never}} = 2 - 0 = 2.$$

The wrong counterfactual trend

Early adopters' change includes a growing treatment effect, not just an untreated trend.

Goodman-Bacon: One Coefficient, Several Comparisons



Goodman-Bacon's decomposition explains why **comparison composition** matters when treatment effects evolve.

A precise distinction

The basic decomposition has non-negative weights on the 2x2 DiDs.
Those DiDs can still subtract changing effects in already-treated units.

Goodman-Bacon (2021), Journal of Econometrics.

Heterogeneous Effects

- Effects may differ across cohorts.
- Effects may differ by exposure length.
- Early adopters may be systematically different.
- Treatment quality may change as the policy scales.

Example

Early minimum-wage adopters may be high-income cities, while later adopters may face different labour-market conditions.

Common Mistakes

Watch for these

- Treating insignificant leads as proof of validity.
- Forgetting which period is omitted.
- Interpreting traditional TWFE leads and lags without checking comparisons.
- Mixing already-treated units into comparisons without interpretation.
- Reporting one TWFE coefficient when dynamics are central.
- Ignoring support for long-run event times.

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Potential Outcomes With Treatment Timing

- $Y_{it}(g)$: potential outcome at time t if unit i first receives treatment in period g.
- $Y_{it}(\infty)$: potential outcome at time t if unit i is never treated.
- $G_i = g$: unit i's observed treatment cohort.
- This notation makes timing part of the causal object.

Why new notation?

Static $Y_i(1)$ and $Y_i(0)$ hide when treatment begins and how long exposure has lasted.

Group-Time ATT

$$ATT(g,t) = E[Y_{it}(g) - Y_{it}(\infty) \mid G_i = g], \qquad t \geq g$$

- g: cohort first treated in period g.
- t: calendar period.
- The target is local to a cohort and date.
- Identification compares cohort g to credible never-treated or not-yet-treated units.

A Clean Control Label Still Needs a Trend Assumption

For cohort g, date $t \geq g$, and baseline $b = g - 1$, choose $C_{g,t} = \{i : G_i > t\}$, or use never-treated units only.

Cohort-specific parallel untreated changes

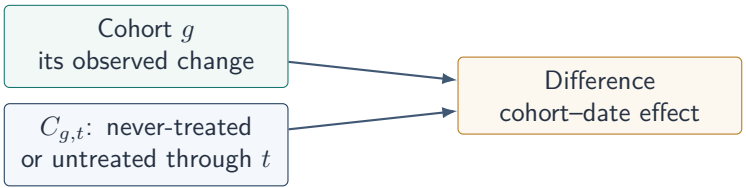
$$E[Y_{it}(\infty) - Y_{ib}(\infty) \mid G_i = g] = E[Y_{it}(\infty) - Y_{ib}(\infty) \mid i \in C_{g,t}].$$

- No anticipation: $Y_{is}(g) = Y_{is}(\infty)$ for $s < g$.
- Controls must remain unaffected at both dates, including by spillovers.
- Observe both changes in comparable populations; require a non-empty control set.

Keep the Familiar Double Difference, Choose Clean Controls

For a fixed cohort g, post date t, and base period $g-1$:

$$ATT(g,t) = E[Y_{it} - Y_{i,g-1} \mid G_i = g] \\ - E[Y_{it} - Y_{i,g-1} \mid i \in C_{g,t}].$$



Requires parallel untreated changes *for this cohort and control set*, no anticipation and no spillovers. Conditional versions also need overlap in X.

Aggregation Is a Policy Choice, Not Just a Software Option

An average at event time k can be written:

$$\theta_k = \sum_{g \in \mathcal{G}_k} w_{g,k}\, ATT(g, g+k), \qquad w_{g,k} \geq 0, \quad \sum_{g \in \mathcal{G}_k} w_{g,k} = 1.$$

- $\mathcal{G}_k$: cohorts with observed outcomes *and valid controls* at that horizon.
- Weights may reflect cohort sizes; state them explicitly.
- Other averages answer cohort, calendar-year or overall policy questions.

A one-minute calculation

At $k = 1$, one cohort has effect 2 and weight 0.75; another has effect 6 and weight 0.25. What does the average describe?

A Long-Run Effect May Describe Only Early Adopters

Suppose observation ends in 2026.

Supported event time	$k = 0$	$k = 1$	$k = 2$	$k = 3$
2023 cohort	Yes	Yes	Yes	Yes
2025 cohort	Yes	Yes	—	—
Number of cohorts	2	2	1	1

A changing population can resemble changing effects

Comparing the average at $k = 0$ with the average at $k = 3$ may change both exposure length and the cohorts represented.

Report support; consider a common cohort set when comparing dynamics.

Inference And Graph Design

- Cluster at the unit or treatment-assignment level.
- Report confidence intervals for event-time or group-time estimates.
- Long leads and lags can be noisy because support falls at the extremes.
- Graphs should show event time, treatment onset, omitted period, uncertainty, comparison group, and estimator.

Graph warning

Do not interpret each dot as equally precise or equally well supported.

R Mini-Lab: One Cohort, Two Dates, Clean Controls

Question: what does the 2021 cohort gain by 2023 in the simulated panel?

```
source("r_exercises/week_06_event_study_staggered.R")
att_gt_one(g = 2021, t = 2023)
with(subset(panel, year == 2023), table(cohort, treated))
```

Five minutes

Name the base year and the valid control cohort. Interpret the estimate in outcome-score units. Why would the 2019 cohort be a problematic untreated control?

These calculations give point estimates; default OLS intervals in the starter are not a complete event-study inference procedure.

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Practice Bridge

- Theory sheet: `homework/session_06_homework.md`.
- QCM: `assessments/qcm/session_06_qcm.md`.
- Exercise: interpret an event-study graph and identify threats.
- Exam skill: explain what a dynamic DiD coefficient compares.

Student deliverable

One paragraph describing leads, lags, omitted period, support, and the relevant comparison group.

Transition To Session 7

- IV identified local effects for compliers.
- Staggered DiD identifies local effects by cohort and calendar time.
- RDD will identify effects local to units around an institutional cutoff.

Bridge intuition

We move from "same trend without treatment" to "smooth potential outcomes at the cutoff".

Key Takeaways

- Event studies show timing, diagnostics, and dynamic effects.
- Leads are useful but not definitive tests.
- Traditional TWFE event studies need caution under staggered adoption and heterogeneous effects.
- Cleaner estimands keep cohort and event time explicit.
- Always report the comparison group and the aggregation.

Appendix: Names To Recognise

- Sun and Abraham: interaction-weighted event studies.
- Callaway and Sant'Anna: group-time average treatment effects.
- Borusyak, Jaravel, and Spiess: imputation approach.
- de Chaisemartin and D'Haultfoeuille: robust DiD estimators.

How to use this appendix

Recognise estimator names in papers; focus class time on the comparison logic and interpretation.

Appendix: A Graph of Intervals Is Not a Joint Test

- A pointwise 95% interval covers one specified horizon with its stated probability.
- A simultaneous band concerns the whole chosen set of coefficients.
- Several individual intervals crossing zero do not establish that all leads are zero.
- Cluster or resample at a level justified by policy assignment and dependence.

Before testing

Use an estimator whose comparisons identify the intended event-time effects.
Better standard errors do not remove contamination from already-treated controls.

Appendix: Two Different Weighting Statements

Basic Goodman-Bacon decomposition:

$$\widehat{\tau}_{TWFE} = \sum_j s_j \widehat{DiD}_j, \qquad s_j \geq 0, \quad \sum_j s_j = 1.$$

Some 2x2 DiDs use already-treated controls and subtract changing effects.

A decomposition into underlying group–time effects:
The implied weights on those causal effects can be negative.

These are different objects. Neither justifies interpreting arbitrary staggered TWFE as a transparent overall ATT.

Goodman-Bacon (2021), Journal of Econometrics.

References: Why Modern DiD Became Necessary

- Goodman-Bacon (2021). "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics*, 225(2), 254–277.
  Published paper.
- Sun and Abraham (2021). "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics*, 225(2), 175–199.
  Author manuscript.
- Callaway and Sant'Anna (2021). "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics*, 225(2), 200–230.
  Author manuscript.